

Module 13: File Handling

13.1 Introduction to Files

- Data stored in variables is lost when the program ends
- Files allow data to persist on disk between program runs
- Types of files
 - Text files: human-readable plain text
 - Binary files: data stored in raw binary format, not human-readable
- Required header: `#include <fstream>`
- Three file stream classes
 - `ofstream`: write to a file
 - `ifstream`: read from a file
 - `fstream`: read and write to a file

13.2 Writing to a File

```
ofstream outFile("data.txt");
if (outFile.is_open()) {
    outFile << "Hello, File!" << endl;
    outFile << 42 << endl;
    outFile.close();
}
```

- File opening modes
 - `ios::out`: create or overwrite (default for `ofstream`)
 - `ios::app`: append to existing content without erasing
 - `ios::trunc`: erase existing content before writing

13.3 Reading from a File

```
ifstream inFile("data.txt");
if (inFile.is_open()) {
    string line;
    while (getline(inFile, line)) {
        cout << line << endl;
    }
    inFile.close();
}
```

- Always check if the file opened successfully before reading or writing
- Use `eof()` or check the stream state in the while condition to detect end of file